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# Modeling of saturated hydraulic conductivity for vegetated soils considering root-soil interactions

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A B S T R A C T

Understanding the interactions between roots and soil provides insight into how plant roots affect soil saturated hydraulic conductivity ($K_{sat}$), playing a key role in maintaining water balance in surface ecosystems. This research focuses on examining the $K_{sat}$ of vegetated soils under varying root densities and orientations. To accurately mimic natural root structures, experimental roots were created using three-dimensional (3D) printing technology. The findings reveal that root-soil interactions primarily involve two processes: the compaction of rhizosphere soil aggregates due to root growth and the creation of channels at the root-soil interface. These interrelated mechanisms significantly influence $K_{sat}$ of soil. Furthermore, new void ratio functions have been introduced to model the changes in void ratio caused by root-soil interactions in saturated soils. To better capture the combined impact of these interactions, the vegetated soil is categorized into two distinct zones: the matrix domain and the preferential flow domain. A dual-permeability model is developed to estimate the $K_{sat}$ of vegetated soils. This work elucidates how root-soil interactions affect $K_{sat}$, contributing to more precise predictions and offering essential scientific support for ecological engineering, water management, and agricultural practices.



## 1. Introduction

Within the complex framework of the Earth's ecosystem, the hydrological cycle occupies a central position, serving not only as a foundation for ecological stability but also exerting a major influence on biodiversity and the delivery of ecosystem services (Weng and Luo, 2008; Gleeson et al., 2012, 2020; Zhao et al., 2021). In this cycle, soil functions as a key reservoir and conduit for water movement, with its hydraulic behaviour closely linked to intrinsic physical attributes such as particle size distribution, dry density, and pore structure. These characteristics directly affect processes like rainwater infiltration, surface runoff formation, and groundwater replenishment (Tamea et al., 2009; Zheng et al., 2019; Cheng et al., 2021; Ng and Peprah-Manu, 2023; Fraccica et al., 2024). Among soil hydraulic properties, $K_{sat}$ stands out as a fundamental indicator of soil permeability to water. It serves as a multidimensional environmental parameter that reflects both the internal architecture of soil and its interactive relationship with vegetation cover in surface ecosystems (Yang et al., 2009; Kiyono et al., 2023; Gholami et al., 2024; Ma et al., 2024). As a result, it has become an essential variable for quantifying the exchange of mass and energy between soil moisture, atmospheric precipitation, and plant systems (Yuliana et al., 2025). Its theoretical and practical importance is evident in the development of ecohydrological models, the evaluation of soil and water conservation effectiveness, and the sustainable management of regional water resources.

The interaction between plant roots and soil particles represents a crucial biophysical process that influences the $K_{sat}$ of soil (Keller et al., 2012; Fischer et al., 2015; Ni et al., 2019; Lei et al.,

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2022; Mair et al., 2022; Qin et al., 2022). As roots extend into the soil, their penetration and growth can significantly modify the bonding conditions and pore structure of the surrounding soil matrix (Lu et al., 2020; Lärm et al., 2024). Research indicates that root biomass development affects how much pore space is occupied by roots, thereby altering soil porosity (Aravena et al., 2011; Cheng et al., 2023; Li et al., 2024). Additionally, expanding roots can apply mechanical stress to adjacent soil aggregates or particles (Zhu et al., 2019; Chen et al., 2022), with recorded pressures reaching as high as 1 MPa (Bengough et al., 2006; Oleghe et al., 2017), which may lead to compaction and reduced porosity in the rhizosphere (Dexter, 1987; Mooney et al., 2012). As a result, soils with vegetation often display lower $K_{sat}$ compared to unvegetated soils (Colombi et al., 2018; Leung et al., 2023). Conversely, other studies have found that root presence and subsequent shrinkage can create preferential flow paths along the root-soil interface (Sullivan et al., 2022; Ni et al., 2024), enhancing pore connectivity and thus increasing $K_{sat}$. These findings emphasize that the formation of the root-soil interface plays a key role in determining the hydraulic properties of vegetated soils (Materechera et al., 1992; Ni et al., 2019; Li et al., 2023). In summary, root-soil interactions exert both positive and negative effects on $K_{sat}$, yet the underlying mechanisms remain insufficiently understood. Moreover, there is a lack of comprehensive theoretical models that effectively account for these interactions. A deeper understanding of root-soil interactions and the development of predictive models are essential to accurately estimate the $K_{sat}$ of vegetated soils and enhance the functionality of vegetated geotechnical systems.

Numerous researchers have developed models to estimate $K_{sat}$ of soil. In the case of bare soil, this property is mainly governed by the void ratio and pore distribution of the soil. Accordingly, many existing models focus on predicting $K_{sat}$ by incorporating factors such as void ratio, pore geometry, and the sinuosity of flow pathways (Carman, 1937; Carrier and Beckman, 1984; Carrier III, 2003). For soils with vegetation, theoretical and numerical approaches have been employed to explore how plant roots influence $K_{sat}$. Utilizing X-ray microtomography and numerical simulations, Aravena et al. (2011, 2014) observed that root-induced compaction enhances the connectivity among soil aggregates, thereby increasing the $K_{sat}$ of soil. Shao et al. (2017) introduced a dual-permeability model integrating two modified Darcy-Richard's equations to represent both the soil matrix and preferential flow domains. The calibrated model was subsequently applied to assess how plant-induced preferential flow affects the stability of vegetated slopes. However, these studies did not fully address the influence of root attributes such as orientation, density, and morphological variations on $K_{sat}$. Ni et al. (2019) explored how root decay and associated volume shrinkage affect $K_{sat}$, proposing a predictive equation for such changes. Their results demonstrated that $K_{sat}$ is highly sensitive to alterations in root structure. Overall, existing models have yet to comprehensively capture the combined effects of root-induced rhizosphere compaction and preferential flow across varying root orientations and densities. Therefore, further research into root-soil interactions under different root distributions and the development of an integrated predictive model remain essential yet challenging tasks.

This research seeks to investigate how interactions between roots and soil influence the $K_{sat}$ of soil and to develop a predictive model for estimating this property in vegetated soils. To facilitate the study, artificial roots are employed as experimental subjects, allowing the investigation of how variations in root density and orientation affect the $K_{sat}$ of soil under different soil dry density conditions. In light of these findings, new void ratio functions are introduced that account for two key processes: the compaction of rhizosphere aggregates caused by root growth and the creation of channels at the root-soil interface within the soil pore structure. Building upon these proposed functions, a dual-permeability model is formulated to capture the combined influence of root-soil interactions on soil pore characteristics, enabling the prediction of $K_{sat}$ in vegetated soils. The developed model is further applied to estimate the $K_{sat}$ of soils containing natural roots. Overall, this study emphasizes the mechanisms through which root-soil interactions affect $K_{sat}$, offering meaningful theoretical support for applications in water resource conservation, agricultural management, and ecological engineering practices.

## 2. Material and methods

### 2.1. Testing material

The soil specimen used in this study, known as Xiashu soil, is a clayey soil commonly found in the middle and lower reaches of the Yangtze River plain in China. Its composition consists of approximately 76 % silt, 22 % clay, and 2 % sand, with a liquid limit of 36.5 % and a plastic limit of 19.5 %. Based on the Standard Proctor compaction test (ASTM D698-12, 2012), the optimal water content for compaction is 16.5 %, at which the soil reaches a maximum dry density of 1.7 Mg/m$^3$. The basic physical properties of the tested soil are summarized in Table 1. According to the Unified Soil Classification System (ASTM D-2487, 2017), Xiashu soil is classified as lean clay (CL).

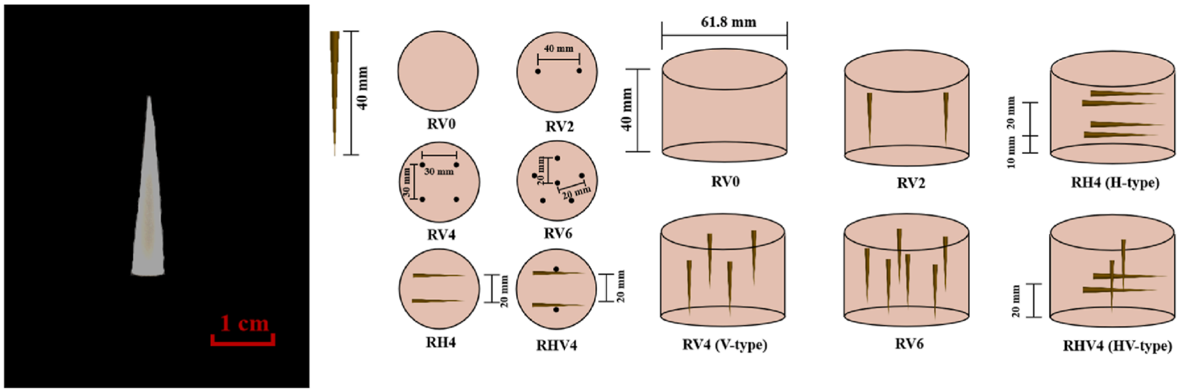
To investigate how roots influence the $K_{sat}$ of soil, this study used three-dimensional (3D) printing technology to create standardized and reproducible roots that can be embedded within the soil matrix. The selected printing material was a photosensitive resin (Wiiboox SP-RH 1001), chosen for its ability to closely mimic the structural and mechanical characteristics, such as stiffness and strength of natural root systems (Chen et al., 2023). In this study, root structures were simplified into a conical form, as shown in Fig. 1. This design choice was based on several considerations: first, many natural root systems, particularly taproots, exhibit a conical shape, gradually tapering from a thicker base to a finer tip. And this morphology provides an efficient combination of tensile strength and material efficiency, enabling effective resistance against soil shear stress (Kamchoom et al., 2014; Kolb et al., 2017). Additionally, the narrowing tip reduces mechanical impedance when penetrating compacted soil layers, facilitating deeper root growth (Lynch et al., 2022). Finally, since most plant species considered in later stages of model validation possess conical root systems (Jotisankasa and Sirirattanachat, 2017; Wang et al., 2020), the artificial roots used in this research were designed accordingly.

### 2.2. Testing program and specimen preparation

The primary focus of this research is to examine how root

**Table 1**
Index properties of Xiashu soil.

| Property | Value |
| --- | --- |
| Specific gravity, $G_s$ | 2.73 |
| Sand content (%) | 2 |
| Slit content (%) | 76 |
| Clay content (%) | 22 |
| Liquid limit, $W_L$ (%) | 36.5 |
| Plastic limit, $W_P$ (%) | 19.5 |
| Plasticity index, $I_p$ (%) | 17.0 |
| Shrinkage limit, $W_s$ (%) | 10.5 |
| Optimum water content, $\omega_{op}$ (%) | 16.5 |
| Maximum dry density (Mg/m$^3$) | 1.7 |




(a) (b)

**Fig. 1.** Morphological characteristics and spatial distribution of artificial roots within soil: (a) Morphological characteristics of artificial roots; and (b) Spatial distribution of artificial roots in soil.

density and orientation influence the $K_{sat}$ of soil. To accomplish this, three sets of hydraulic conductivity tests were conducted on soil specimens under controlled conditions. All test specimens were prepared in cylindrical form, with a diameter of 61.8 mm and a height of 40 mm. Initially, the $K_{sat}$ of bare soil specimens was measured across four different dry density levels: 1.3 Mg/m$^3$, 1.4 Mg/m$^3$, 1.5 Mg/m$^3$, and 1.6 Mg/m$^3$. In the second phase, three distinct root densities were tested at each soil density level. Root density was adjusted by varying the number of artificial roots per specimen, specifically 2, 4, and 6 roots. Root density is defined as the volume of roots per cubic meter of soil, so specimens containing 2, 4, and 6 roots corresponded to root densities of 0.004 m$^3$/m$^3$, 0.009 m$^3$/m$^3$, and 0.013 m$^3$/m$^3$, respectively. In this study, root density serves as an indicator of average root spacing (Huo et al., 2021; Meijer, 2021). Naturally, plant root systems typically occupy between 0.3 % and 3 % of the total soil volume (Jackson et al., 1996; Gregory, 2006), and the selected root density range aligns well with this natural variation. In the final series of experiments, three root orientations were evaluated under uniform soil dry density conditions: horizontal (H-type), vertical (V-type), and combined horizontal-vertical (HV-type). Root orientation refers to the dominant direction and spatial arrangement of root growth within the soil matrix (Mair et al., 2022). According to Wang et al. (2020), root orientations can be classified into six categories: horizontal (H), horizontal-vertical (HV), vertical (V), mixed (M), radial (R), and winding (W). This study specifically examined H, HV, and V types. The parameters of the experimental setup are presented in Table 2. For identification purposes, specimen labels such as “BV0D13” and “RH4D13” were used. In this labelling system, “B” indicates bare soil while “R” denotes vegetated soil; “V” and “H” represent vertical and horizontal root orientations, respectively. The numbers “0” and “4” specify the quantity of roots in the specimen, and “D13” signifies a soil dry density of 1.3 Mg/m$^3$.

### 2.3. Testing apparatus and procedure

The $K_{sat}$ of the soil specimens was assessed using a falling head permeability test in accordance with ASTM D-1586 (2008). The experimental setup is illustrated in Fig. 2. For all tests conducted in this study, the soil specimens were subjected to unidirectional vertical seepage, effectively excluding any influence from lateral water movement. A constant water supply was maintained through a water bottle connected to the system. To minimize air entrapment and ensure accurate measurements, de-aired water was introduced into the permeameter from the bottom upward (Cuisinier et al., 2011). For each specimen, nine initial hydraulic head measurements were recorded and averaged to determine the saturated hydraulic conductivity. The calculation method is based on the following formula:

$$K_{sat} = 2.3\frac{aL}{A(t_2 - t_1)}\log_{10}\frac{H_1}{H_2} \tag{1}$$

where $K_{sat}$ represents the saturated hydraulic conductivity (m/s); $L$ refers to the height of the soil specimen within the cutting ring (m); $A$ indicates the cross-sectional area of the specimen (m$^2$); $a$ denotes the cross-sectional area of the variable head tube (m$^2$); $H_1$ and $H_2$ are the hydraulic head levels at the beginning and end of the test, respectively (m); and $t_1$ and $t_2$ correspond to the initial and final time points of the test, respectively (s).

## 3. Experimental results and discussion

### 3.1. $K_{sat}$ of vegetated soils incorporating root densities

Fig. 3a presents the variation in $K_{sat}$ of vegetated soil specimens with respect to root density under varying soil dry density conditions. For specimens containing vertically oriented roots and at soil dry densities of 1.4 Mg/m$^3$ and 1.5 Mg/m$^3$, the $K_{sat}$ increases as root density increases. More specifically, when root density rose from 0 to 0.013 m$^3$/m$^3$, the $K_{sat}$ increased by 180 % (from 1.0 × 10$^{-6}$ m/s to 2.8 × 10$^{-6}$ m/s) for a dry density of 1.4 Mg/m$^3$, and by 75 % (from 0.4 × 10$^{-6}$ m/s to 0.7 × 10$^{-6}$ m/s) for a dry density of 1.5 Mg/m$^3$. As shown in the figure, for soil dry densities of 1.3 Mg/m$^3$ and 1.6 Mg/m$^3$, the trend in $K_{sat}$ initially rises and then declines with increasing root density from 0 to 0.013 m$^3$/m$^3$. The peak values occur at a root density of 0.009 m$^3$/m$^3$, where the corresponding $K_{sat}$ were measured at 4.0 × 10$^{-6}$ m/s (for 1.3 Mg/m$^3$) and 0.2 × 10$^{-6}$ m/s (for 1.6 Mg/m$^3$). A notable distinction is that when root density exceeds 0.004 m$^3$/m$^3$, the $K_{sat}$ of the 1.6 Mg/m$^3$ dry density specimens drop below that of bare soil, with a maximum reduction of 45.2 %. In contrast, for the 1.3 Mg/m$^3$ dry density condition, the $K_{sat}$ of vegetated soil remains consistently higher than that of bare soil.

Plant roots have a significant influence on the $K_{sat}$ of vegetated soil, which is primarily driven by two root-soil interaction mechanisms: the compaction of rhizosphere soil aggregates and the development of root-induced preferential flow channels (Ng et al., 2022; Li et al., 2023). Particularly, when the soil dry density is

**Table 2**
Details of the saturated hydraulic conductivity test based on artificial roots.

| Influence factor | Test ID | Number of artificial roots in the test (pcs) | Soil dry density (Mg/m$^3$) | Root orientation | Root density (m$^3$/m$^3$) |
|---|---|---|---|---|---|
| Dry density effects on $K_{sat}$ of bare soil | BV0D13 | 0 | 1.3 | – | 0 |
| | BV0D14 | | 1.4 | | |
| | BV0D15 | | 1.5 | | |
| | BV0D16 | | 1.6 | | |
| Root density effects on $K_{sat}$ of vegetated soils | RV2D13 | 2 | 1.3 | V-type | 0.004 |
| | RV2D14 | | 1.4 | | |
| | RV2D15 | | 1.5 | | |
| | RV2D16 | | 1.6 | | |
| | RV4D13 | 4 | 1.3 | V-type | 0.009 |
| | RV4D14 | | 1.4 | | |
| | RV4D15 | | 1.5 | | |
| | RV4D16 | | 1.6 | | |
| | RV6D13 | 6 | 1.3 | V-type | 0.013 |
| | RV6D14 | | 1.4 | | |
| | RV6D15 | | 1.5 | | |
| | RV6D16 | | 1.6 | | |
| Root orientation effects on $K_{sat}$ of vegetated soils | RH4D13 | 4 | 1.3 | H-type | 0.009 |
| | RH4D14 | | 1.4 | | |
| | RH4D15 | | 1.5 | | |
| | RH4D16 | | 1.6 | | |
| | RHV4D13 | 4 | 1.3 | HV-type | 0.013 |
| | RHV4D14 | | 1.4 | | |
| | RHV4D15 | | 1.5 | | |
| | RHV4D16 | | 1.6 | | |

Note: "pcs" is a unit of quantity indicating the number of artificial roots placed in each specimen.

below 1.6 Mg/m$^3$, the formation of preferential flow paths becomes more evident, leading to an increase in the $K_{sat}$ of vegetated soil compared to bare soil (Ng et al., 2014; Liu et al., 2018; Feng et al., 2019). Furthermore, at soil dry densities of 1.4 Mg/m$^3$ and 1.5 Mg/m$^3$, increasing root density generates more macropores around the roots, facilitating the development of additional preferential flow channels and further enhancing $K_{sat}$ (Ng et al., 2020). However, at a soil dry density of 1.3 Mg/m$^3$, the situation changes as root density exceeds 0.009 m$^3$/m$^3$. In this case, localized compaction caused by densely packed roots leads to a decline in $K_{sat}$. Nevertheless, this reduction does not outweigh the positive impact of preferential flow, so the $K_{sat}$ of vegetated soil remains

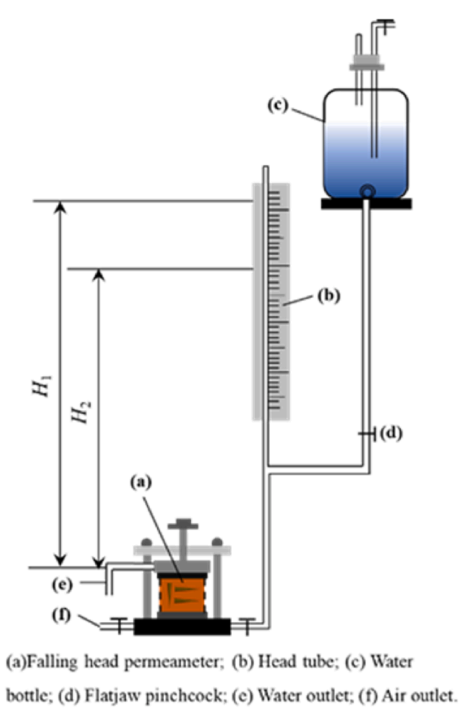



**Fig. 2.** Schematic diagram of the experimental set-up.

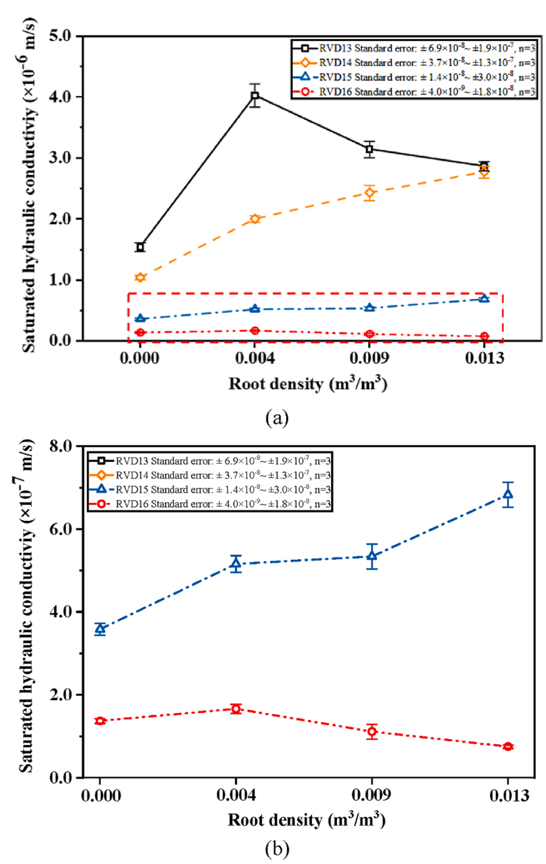



**Fig. 3.** Effects of root density on $K_{sat}$: (a) Root density; and (b) Enlarged local image (1.5 Mg/m$^3$ and 1.6 Mg/m$^3$).

higher than that of bare soil. In contrast, at a soil dry density of 1.6 Mg/m$^3$, the soil matrix is already relatively compact. As root density increases, the compaction effect within the rhizosphere becomes more pronounced. When root density surpasses 0.009 m$^3$/m$^3$, this compaction dominates over the formation of preferential pathways, resulting in a $K_{sat}$ that is lower than that of bare soil (Lucas, 2022).

Fig. 4a shows the influence of soil dry density on the $K_{sat}$ of specimens with varying root densities. For a given root density, an increase in soil dry density leads to a decrease in $K_{sat}$. In the case of bare soil and specimens with a root density of 0.009 m$^3$/m$^3$, the $K_{sat}$ initially drops sharply (when soil dry density is below 1.5 Mg/m$^3$), followed by a more gradual decline, resulting in overall reductions of 91 % and 96 %, respectively. For specimens with root densities of 0.009 m$^3$/m$^3$ and 0.013 m$^3$/m$^3$, the trend in $K_{sat}$ shows an initial slow decrease, followed by a rapid drop, and then a further deceleration. Specifically, when the root density is 0.009 m$^3$/m$^3$, the $K_{sat}$ decreases significantly by $1.9 \times 10^{-6}$ m/s as the soil dry density increases from 1.4 Mg/m$^3$ to 1.5 Mg/m$^3$. In comparison, the reduction is only $0.7 \times 10^{-6}$ m/s when dry density increases from 1.3 Mg/m$^3$ to 1.4 Mg/m$^3$ and $0.4 \times 10^{-6}$ m/s when it rises from 1.5 Mg/m$^3$ to 1.6 Mg/m$^3$. Furthermore, for specimens with a root density of 0.013 m$^3$/m$^3$, the $K_{sat}$ remains nearly constant at approximately $2.8 \times 10^{-6}$ m/s as soil dry density increases from 1.3 Mg/m$^3$ to 1.4 Mg/m$^3$. However, when dry density increases from 1.5 Mg/m$^3$ to 1.6 Mg/m$^3$, the decrease reaches $0.6 \times 10^{-6}$ m/s, while the largest drop of $2.1 \times 10^{-6}$ m/s occurs between 1.4 Mg/m$^3$ and 1.5 Mg/m$^3$. In summary, both root density and soil dry density play critical roles in determining the $K_{sat}$ of soil. As soil dry density increases, pore space diminishes, which directly reduces $K_{sat}$. Roots influence this property through two primary mechanisms: compaction of rhizosphere aggregates, which increases local soil density, and the formation of preferential flow pathways induced by root structures (Lann et al., 2024). These mechanisms have contrasting effects—rhizosphere compaction tends to reduce $K_{sat}$, whereas preferential flow enhances it. The dominant mechanism determines whether the $K_{sat}$ of vegetated soil is higher or lower than that of bare soil. Experimental findings indicate that in loosely packed soils (dry density <1.5 Mg/m$^3$), root-induced preferential flow dominates, leading to increased $K_{sat}$ compared to bare soil. Conversely, in denser soils (dry density ≥1.5 Mg/m$^3$), rhizosphere compaction becomes more pronounced with increasing root density, ultimately causing the $K_{sat}$ of vegetated soil to fall below that of bare soil.

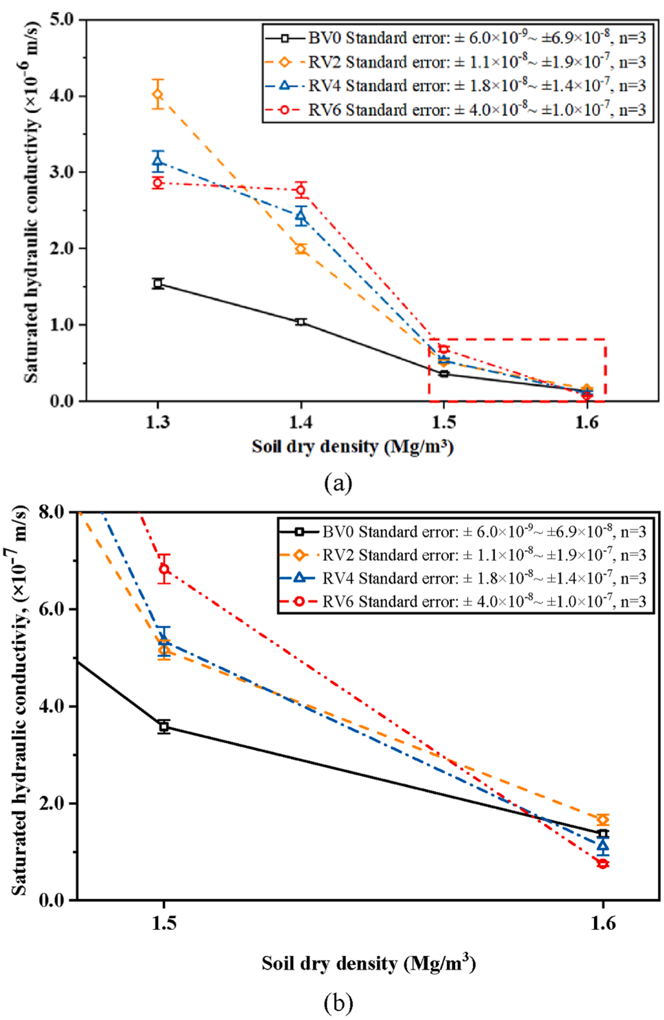



**Fig. 4.** Effects of soil dry density on $K_{sat}$: (a) Soil dry density; and (b) Enlarged local image (from 1.5 Mg/m$^3$ to 1.6 Mg/m$^3$).

## 3.2. $K_{sat}$ of vegetated soils incorporating root orientations

In addition to root density, the orientation of roots also plays a crucial role in shaping soil structure and influencing $K_{sat}$. Fig. 5 demonstrates how different root orientations affect $K_{sat}$ across various soil dry densities. It can be observed that, under the same soil dry density conditions, the highest $K_{sat}$ occurs in soils with vertically oriented (V-type) roots, followed by those with combined horizontal-vertical (HV-type) roots, while horizontally oriented (H-type) roots result in the lowest values. This trend is attributed to the fact that horizontal roots tend to disrupt the natural vertical flow of water and create partial obstructions, unlike their vertical counterparts (Caldwell et al., 1998). As a result, H-type root systems generally lead to lower $K_{sat}$ compared to V-type roots. Moreover, for soil dry densities below 1.5 Mg/m$^3$, the $K_{sat}$ of vegetated soils consistently surpasses that of bare soil, regardless of root orientation, with an observed maximum increase of 134 %. However, at a soil dry density of 1.5 Mg/m$^3$, only specimens with H-type roots exhibit a $K_{sat}$ lower than that of bare soil, showing a reduction of about 10 %. When soil dry density reaches 1.6 Mg/m$^3$, all three types of root orientations yield lower $K_{sat}$ than bare soil, with the most significant decrease reaching 32 %. These observations indicate that two primary root-soil interactions (root-induced preferential flow and rhizosphere aggregate compaction) continue to occur regardless of root orientation. The relative dominance of these mechanisms shifts with changes in soil dry density. Specifically, the influence of

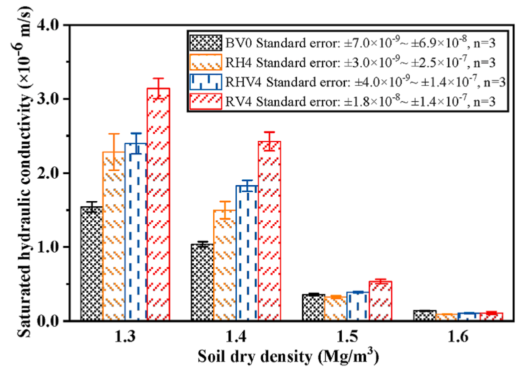



**Fig. 5.** Effects of root orientations on $K_{sat}$ with varying soil dry densities.

preferential flow is more evident in soils with dry densities below 1.5 $Mg/m^3$. As soil becomes denser, the compaction effect within the rhizosphere gradually prevails, leading to reduced $K_{sat}$ in vegetated soils compared to bare soil at dry densities of 1.5 $Mg/m^3$ and 1.6 $Mg/m^3$. In conclusion, even when root orientation varies, roots continue to induce both compaction in the rhizosphere and the formation of preferential flow pathways. However, changes in root orientation can change the direction and continuity of water movement through the soil matrix, thereby modifying flow paths and affecting overall $K_{sat}$ (Angers and Caron, 1998).

Studying how root-soil interactions affect $K_{sat}$ of soil carries significant practical value in engineering. Firstly, roots contribute to the compaction of soil aggregates, increasing the overall matrix density and forming localized zones of higher compaction (Aravena et al., 2011). This enhancement leads to improved mechanical stability, faster rhizosphere consolidation, reduced settlement risks, and restricted radial movement of contaminants (Bengough et al., 2011; Scholl et al., 2014; Shi et al., 2023; Liu et al., 2024). Such properties can be strategically applied in slope stabilization, soft ground improvement, and the development of natural pollutant containment barriers. Conversely, the spaces formed at the root-soil interface act as preferential channels for water movement, promoting the development of interconnected flow networks that support microbial activity (Stokes et al., 2014). These pathways offer promising applications in urban hydrology by facilitating rapid rainwater infiltration, thereby addressing urban flooding issues associated with the “sponge city” concept. Additionally, they are beneficial for ecological recovery in drought environments. By enabling swift downward transport of surface water, these channels enhance soil water retention, improve gas exchange within the soil profile, and promote microbial processes essential for nutrient cycling (Stokes et al., 2014; Kolb et al., 2017).

## 4. Modelling of $K_{sat}$ of vegetated soils

### 4.1. Model formulation

As presented in Fig. 6, the saturated soil containing artificial roots comprises not only the conventional pore water and solid phases (soil particles) but also two root-related components: the root material itself and the root-soil interface channel. These elements can be categorized into two distinct domains: the matrix domain, consisting of pore water and soil particles, and the preferential flow domain, which includes the roots and their associated interface channels. The void ratio within the matrix domain ($e_m$) can be described as follows:

$$e_m = \frac{P_v}{S_v + R_v} \tag{2}$$

where $P_v$, $S_v$, and $R_v$ are the volume ratios of pore water, solid, and root, which are defined as the volumetric ratios of pore water, solid, and root within a unit volume of soil; $R_v$ is quantitatively equivalent to the root density. The formulations for $P_v$ and $S_v$ are as follows:

$$P_v = \frac{e_0}{1+e_0} - (1+\beta)R_v \tag{3}$$

$$S_v = \frac{1}{1+e_0} \tag{4}$$

where $e_0$ signifies the initial bare soil void ratio, and $g$ is the ratio of root-soil interface channel volume to root volume. While the void ratio of the preferential flow domain ($e_f$) may be quantified as:

$$e_f = \frac{C_v}{S_v + R_v} \tag{5}$$

$$C_v = \beta R_v \tag{6}$$

where $C_v$ is the volumetric ratio of root-soil interface channel per unit volume of soil.

In field investigations, two critical parameters, i.e. pore-water volume ratio ($P_v$) and root volume ratio ($R_v$), can be efficiently obtained through sampling. Under controlled laboratory conditions, the pore-water volume ratio ($P_v$) can be quickly assessed using straightforward techniques such as the wax sealing method. Additionally, plant roots extracted from natural soil environments can be analysed using established root image analysis software (WinRHIZO) to accurately determine the root volume ratio ($R_v$) (Wang et al., 2020). With these two parameters, it becomes feasible to separately calculate the void ratios of both the matrix domain and the preferential flow domain. These derived values can then be integrated into the proposed predictive model to estimate the $K_{sat}$ of vegetated soils. Compared to conventional in-situ testing methods such as single-ring or double-ring infiltration tests, this approach eliminates the need for extensive site preparation and equipment setup. Furthermore, it avoids the labor-intensive process of continuous water level monitoring, which often spans several hours or even days (Angulo-Jaramillo et al., 2000). As a result, this method significantly reduces both time consumption and operational effort, thereby improving overall testing efficiency.

Root-soil interactions have a clear influence on soil structure. Roots apply mechanical pressure on adjacent soil particles, leading to a reduction in the void ratio of the matrix domain (Aravena et al., 2011; Bengough et al., 2011). Simultaneously, the development of root-soil interface channels contributes significantly to the formation of preferential flow paths. Lascurain et al. (2021) reported increased porosity around plant roots, resulting in the creation of gaps at the root-soil interface—a phenomenon consistent with the interface channels described in this study. Therefore, modelling the impact of root compaction and root-induced channels on rhizosphere porosity by incorporating additional pore space fractions $\beta$ and $R_v$ offers a simplified yet scientifically sound approach, as shown in Fig. 6. When $R_v$ equals zero, it indicates the absence of roots in the soil. The maximum possible value of $R_v$ is constrained by $e_0/(1+e_0)$, representing complete root occupation of available pore space. Determining the value of $\beta$ requires further investigation into the biomechanical behaviour of specific plant species and the physical properties of the surrounding soil.

To quantify the influence of roots on soil $K_{sat}$, an equation based on soil void ratio was employed to estimate the $K_{sat}$ of bare soil. The Kozney-Carman (KC) equation (Carman, 1937; Carrier III, 2003) has been widely used for calculating $K_{sat}$. Although originally developed for coarse-grained soils, subsequent research has confirmed its applicability across various soil types, including fine-grained clays. It is now recognized that the KC model's utility in estimating $K_{sat}$ and specific surface area extends to the boundaries defined by Darcy's law when appropriate parameter adjustments are applied. The validity of the KC model is supported when the $K_{sat}$ falls within the range of $10^{-12}$-$10^{-2}$ m/s (Chapuis and Aubertin, 2003; Urumović and Urumović Sr, 2016). In this study, the measured $K_{sat}$ ranged between $10^{-7}$ m/s and $10^{-4}$ m/s, which lies well within the acceptable application range of the model.

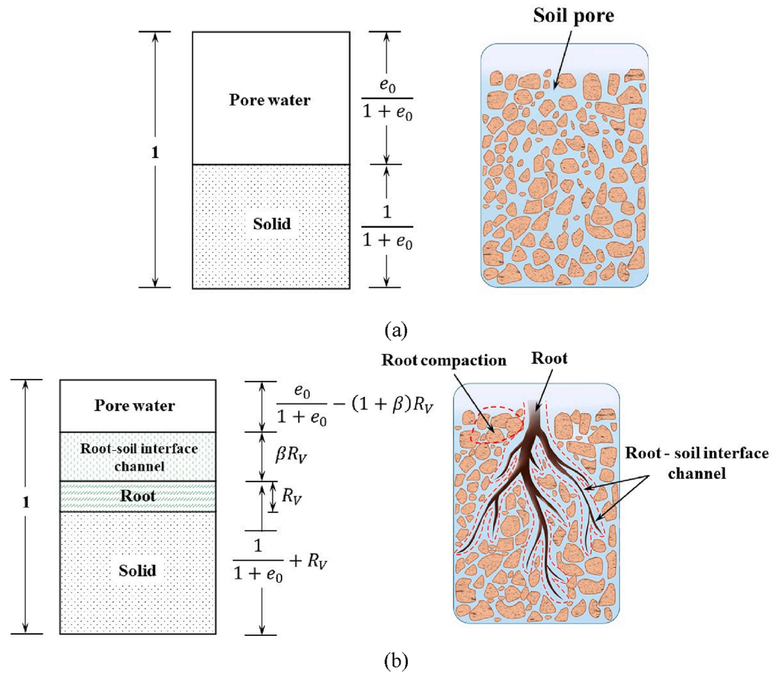



**Fig. 6.** Phase diagram of saturated soil with vegetation considering root compaction and root-soil interface channels: (a) Bare soil; and (b) Vegetated soil.

The mathematical expression for the equation is as follows:

$$k_B = \mu \frac{e^3}{(1+e)} \tag{7}$$

where $k_B$ denotes the $K_{sat}$ of bare soil; $e$ refers to the soil's void ratio; and $\mu$ is a coefficient that incorporates various factors, such as pore geometry, flow path tortuosity, specific surface area of soil particles in contact with water, water viscosity, and the unit weight of water.

The total $K_{sat}$ of vegetated soil can be expressed as:

$$k_R = k_m + k_f = \mu_m \frac{{e_m}^3}{(1+e_m)} + \mu_f \frac{e_f^3}{\left(1+e_f\right)} \tag{8}$$

where the subscripts "m" and "f" denote the matrix domain and the preferential flow domain, respectively. $k_m$ and $k_f$ correspond to the $K_{sat}$ of the matrix domain and the rhizosphere domain, respectively. The void ratios are adjusted using Eqs. (2) and (3) can be incorporated into the KC equation to calculate the $K_{sat}$ of the matrix and rhizosphere domains. By integrating the values of $k_m$ and $k_f$, a predictive model for the $K_{sat}$ of vegetated soil is established, which considers both the compaction of rhizosphere soil aggregates induced by roots and the presence of root-soil interface channels.

To capture the effect of roots, the parameter $\mu$ in Eq. (4) can be modified as follows:

$$\mu_m = C \frac{\mu}{\exp\,(\eta R_v)} \tag{9}$$

$$\mu_f = C \frac{\mu}{\exp\left(-\eta \beta^2\right)} \tag{10}$$

where $C$ is a parameter influenced by root orientation, with values typically ranging between 0.7 and 1.0. According to experimental measurements of $K_{sat}$ in specimens containing artificial roots, the conductivity for H-type and HV-type roots is approximately 0.7 and 0.8 times that of V-type roots, respectively. This variation is closely linked to the proportion of horizontally oriented roots: higher horizontal root content results in $C$ values closer to 0.7. The precise value of $C$ can be calculated using linear interpolation based on the percentage of horizontally distributed roots. Therefore, $C$ equals 0.7 when all roots are H-type, 1.0 when all roots are V-type, and takes on an intermediate value when the root system includes both orientations (HV-type). As the proportion of vertical roots increases, the value of $C$ approaches 1.0. The parameter $\eta$ accounts for pore geometry and flow path tortuosity, which are affected by both soil characteristics and plant species. Overall, when $R_v = 0$, indicating no plant roots are present, the preferential flow void ratio $e_f$ also equals zero, leading $\mu_m$ to equal $\mu$. As a result, the computed $k_R$ corresponds to the $K_{sat}$ of bare soil, confirming that Eq. (8) is equally valid for bare soil conditions.

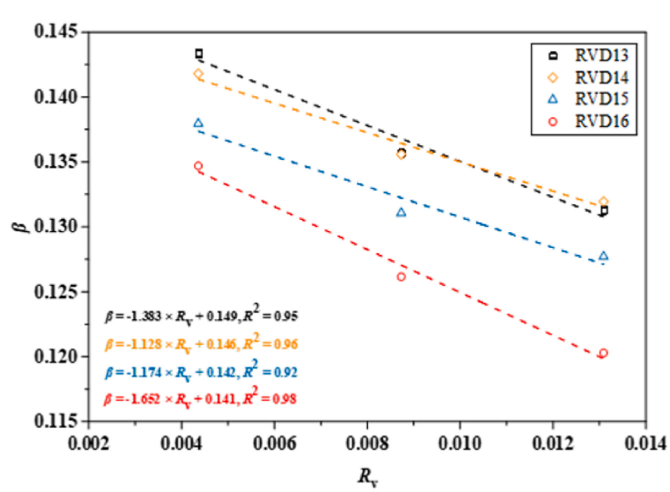



**Fig. 7.** The relationship between $\beta$ and $R_v$.

It can be shown in Fig. 7, the relationship between $\beta$ and $R_v$ follows a linear relationship, which can be expressed as:

$$\beta = \alpha R_v + m \tag{11}$$

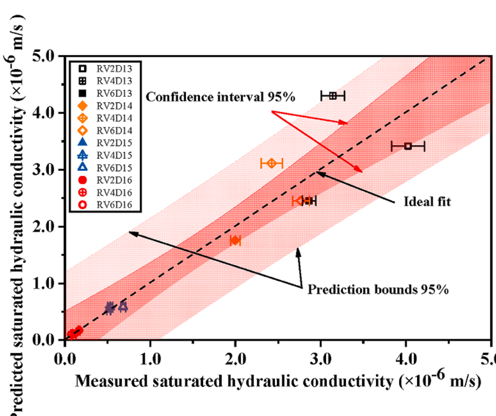
where the fitting parameters $\alpha$ and $m$ depend on soil properties and plant species.

Most current models tend to neglect the combined effects of root-soil interactions, particularly the simultaneous influence of root compaction and root-soil interface channels, on $K_{sat}$ of soil. Instead, they typically address either the displacement of soil pores by root volume or the development of preferential flow paths at the root-soil interface (Aravena et al., 2011; Shao et al., 2017). In contrast, the model introduced in this study innovatively categorizes vegetated soil into two distinct regions: the matrix domain and the preferential flow domain. Within the matrix domain, the reduction in soil porosity caused by root volume and compaction is considered, while in the preferential flow domain, the contribution of root-soil interface channels to enhanced water flow is evaluated. Following this classification, an improved version of the KC equation is applied to separately estimate the $K_{sat}$ of each domain. These results are then integrated to develop a comprehensive predictive model that accounts for the dual influence of root-soil interactions on soil hydraulic properties. Moreover, unlike many existing models that are limited by specific plant species (Ni et al., 2019), the proposed approach depends only on the volumetric distribution and spatial orientation of roots within the soil, making it independent of vegetation type and thus more broadly applicable across different ecological and engineering contexts.

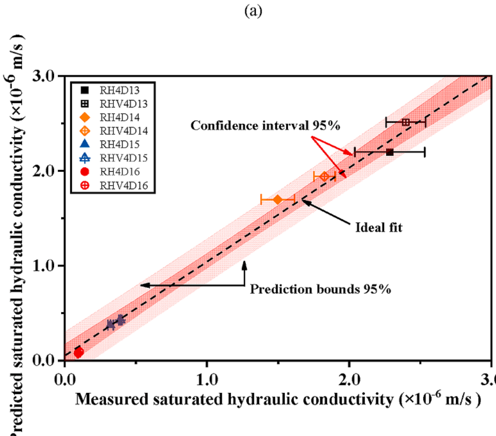
### 4.2. Model validation

Fig. 7 compares the $K_{sat}$ of vegetated soil containing artificial roots, as predicted by the proposed model under varying root densities and orientations, with the corresponding experimental measurements. The predicted and observed values are plotted together, along with the 1:1 ideal fit line (contour line), representing perfect agreement. Additionally, both the 95 % confidence interval and the 95 % prediction bounds are displayed in the figure. For different root densities and orientations, the standard errors between predicted and measured values are 0.14 and 0.03, respectively, with a coefficient of determination ($R^2$) exceeding 0.89 for both datasets. As shown in Fig. 8a, nearly all data points lie close to the ideal fit line, indicating strong alignment between predictions and measurements. In Fig. 8b, almost all data points fall within the 95 % confidence interval. Furthermore, the width of both the confidence and prediction intervals is narrower than in Fig. 8a, with a reduction in standard error by approximately 0.11. The maximum standard error remains below 0.14. Considering the limited dataset used in this study and the practical application of the model in predicting the $K_{sat}$ of vegetated soils, the achieved level of accuracy is considered satisfactory.

The model's performance in predicting the $K_{sat}$ of soil with real plant roots is presented in Fig. 9. Specifically, Fig. 9a presents the predicted values for soil specimens vegetated with ryegrass, based on the proposed model, across various root densities and soil dry

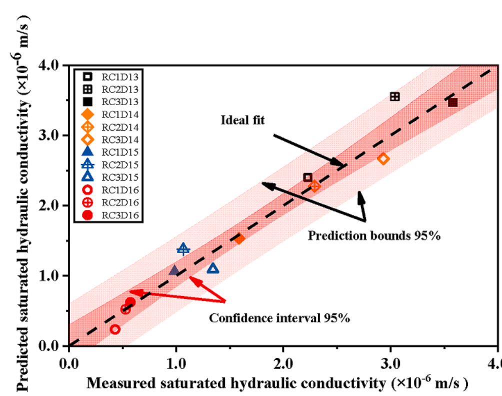



(a)

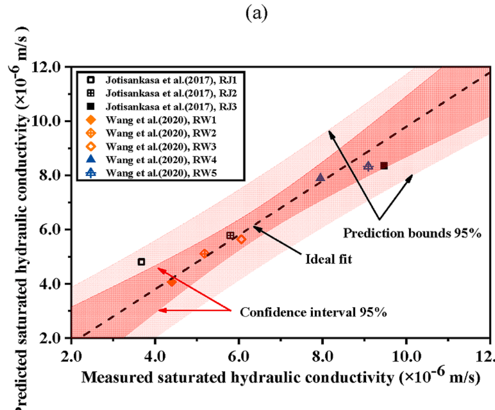



(b)

**Fig. 8.** Performance evaluation of the model for predicting the $K_{sat}$ of vegetated soil with artificial roots: (a) Different root densities, and (b) Different root orientations.




(a)




(b)

**Fig. 9.** Performance evaluation of model for predicting the $K_{sat}$ of vegetated soil with authentic roots: (a) Effects of varying root densities and soil dry densities; and (b) Effects of species of authentic plant roots.

densities. The experimental data were sourced from Cui et al. (2024). According to their study, ryegrass roots primarily developed vertically with minimal lateral spread, leading to the assignment of a root orientation factor ($C$) of 1.0. Despite variations in root density and soil dry density, the predicted and measured values show a high degree of consistency. Most data points cluster closely around the contour line, with an $R^2$ value greater than 0.95. Fig. 9b displays the predicted $K_{sat}$ for multiple plant species with natural root systems, using experimental data from Jotisankasa and Sirirattanachat (2017) and Wang et al. (2020). Many herbaceous plants exhibit predominantly vertical root growth with limited horizontal development, which corresponds to V-type root orientation. The model was also applied to predict the $K_{sat}$ of woody plant-root systems under three root orientation patterns: H-type, HV-type, and V-type. Jotisankasa and Sirirattanachat (2017) studied the influence of Vetiver grass roots on soil hydraulic properties, while Wang et al. (2020) investigated the effects of various woody species. In both cases, most data points align closely with the ideal fit line, with a standard error of 0.23 and an $R^2$ value exceeding 0.85 for both predicted and measured results. The parameter values used in the model for these experiments are summarized in Tables 3 and 4. Overall, the model demonstrates reliable predictive capability for soils containing both herbaceous and woody plant roots, highlighting its broad applicability and generalization potential.

Furthermore, the proposed model indicates considerable potential for meaningful future applications. For example, by integrating predictions of extreme rainfall events under projected climate change scenarios, the model can assist in selecting appropriate vegetation types that optimize both water infiltration and erosion control. This capability is particularly valuable in floodplain management, where strategic vegetation choices can significantly improve flood mitigation capacity. Moreover, the impact of root-soil interactions on $K_{sat}$ is closely tied to the dynamics of the soil carbon cycle. As a result, the model's parameters can be used as indirect metrics for evaluating the potential of soils to sequester carbon (Domec et al., 2010). This dual functionality highlights the broader environmental relevance of the model beyond hydrological applications.

**Table 3**
Values of the parameters in the model (Artificial roots).

| Influence factor | Test ID | $\mu$ (m/s) | $\alpha$ | $m$ | $\eta$ | $R_v$ | $C$ | $\beta$ |
|---|---|---|---|---|---|---|---|---|
| Effects of varying root densities | RV2D13 | 2.153 × 10-6 | −1.383 | 0.149 | 1000 | 0.004 | 1.00 | 0.143 |
| | RV4D13 | | | | | 0.009 | | 0.137 |
| | RV6D13 | | | | | 0.013 | | 0.131 |
| | RV2D14 | 2.153 × 10-6 | −1.128 | 0.146 | | 0.004 | | 0. 141 |
| | RV4D14 | | | | | 0.009 | 1.00 | 0.136 |
| | RV6D14 | | | | | 0.013 | | 0.132 |
| | RV2D15 | 2.153 × 10-6 | −1.174 | 0.142 | | 0.004 | | 0.137 |
| | RV4D15 | | | | | 0.009 | | 0.132 |
| | RV6D15 | | | | | 0.013 | 1.00 | 0.127 |
| | RV2D16 | 2.153 × 10-6 | −1.652 | 0.141 | | 0.004 | | 0.134 |
| | RV4D16 | | | | | 0.009 | | 0.127 |
| | RV6D16 | | | | | 0.013 | | 0.120 |
| Effects of varying root orientations | RH4D13 | 2.153 × 10-6 | – | – | 1000 | 0.009 | 0.70 | 0.137 |
| | RH4D14 | | | | | | | 0.136 |
| | RH4D15 | | | | | | | 0.132 |
| | RH4D16 | | | | | | | 0.127 |
| | RHV4D13 | 2.153 × 10-6 | – | – | | 0.009 | 0.80 | 0.137 |
| | RHV4D14 | | | | | | | 0.136 |
| | RHV4D15 | | | | | | | 0.132 |
| | RHV4D16 | | | | | | | 0.127 |

**Table 4**
Values of the parameters in the model (Authentic roots).

| Influence factor | Test ID | $\mu$ (m/s) | $\alpha$ | $m$ | $\eta$ | $R_v$ | $C$ | $\beta$ |
|---|---|---|---|---|---|---|---|---|
| Effects of varying root densities and soil dry densities | RC1D13 | 2.153 × 10-6 | −0.895 | 0.144 | 1000 | 0.00518 | 1.00 | 0.140 |
| | RC2D13 | | | | | 0.01453 | | 0.131 |
| | RC3D13 | | | | | 0.01752 | | 0.129 |
| | RC1D14 | 2.153 × 10-6 | −0.808 | 0.143 | | 0.00587 | | 0. 138 |
| | RC2D14 | | | | | 0.00906 | 1.00 | 0.135 |
| | RC3D14 | | | | | 0.01418 | | 0.131 |
| | RC1D15 | 2.153 × 10-6 | −1.450 | 0.146 | | 0.00409 | | 0.141 |
| | RC2D15 | | | | | 0.00671 | | 0.137 |
| | RC3D15 | | | | | 0.00835 | 1.00 | 0.134 |
| | RC1D16 | 2.153 × 10-6 | −1.106 | 0.142 | | 0.00313 | | 0.139 |
| | RC2D16 | | | | | 0.00569 | | 0.136 |
| | RC3D16 | | | | | 0.0736 | | 0.134 |
| Effects of species of authentic plant roots | RJ1 | 1.458 × 10-5 | −0.783 | 0.188 | 500 | 0.00856 | 1.00 | 0.181 |
| | RJ2 | | | | | 0.00913 | | 0.181 |
| | RJ3 | | | | | 0.01356 | | 0.177 |
| | RW1 | 2.167 × 10-6 | −31.331 | 0.501 | 100 | 0.00050 | 0.70 | 0.485 |
| | RW2 | | | | | 0.00056 | 0.80 | 0.483 |
| | RW3 | | | | | 0.00060 | 0.80 | 0.482 |
| | RW4 | | | | | 0.00072 | 1.00 | 0.478 |
| | RW5 | | | | | 0.00079 | 1.00 | 0.476 |

## 5. Conclusions

This research investigates how root-soil interactions affect the $K_{sat}$ of vegetated soils. Soil specimens with varying root densities and orientations were analysed to evaluate their impact on hydraulic properties. Furthermore, new soil void ratio formulations and a predictive model for estimating the $K_{sat}$ of vegetated soils are introduced. The proposed model accounts for two key mechanisms: root-induced compaction of rhizosphere aggregates and the formation of preferential flow paths at the root-soil interface. Based on the findings, the following conclusions are drawn.

(1) The effect of roots on $K_{sat}$ is dependent on soil dry density. When the soil dry density is below 1.6 Mg/m$^3$, root presence enhances $K_{sat}$. However, this trend reverses at a density of 1.6 Mg/m$^3$. For soil dry densities of 1.4 Mg/m$^3$ and 1.5 Mg/m$^3$, increasing root density leads to higher $K_{sat}$, up to 167 % and 91 %, respectively, compared to bare soil. In contrast, at a density of 1.6 Mg/m$^3$, a decreasing trend in $K_{sat}$ is observed with increasing root density. At a lower density of 1.3 Mg/m$^3$, $K_{sat}$ initially rises and then decreases with greater root distribution, yet remains at least 86 % higher than that of bare soil.

(2) Roots oriented horizontally tend to restrict vertical water movement. Among different root orientations, vertically oriented roots (V-type) result in the highest $K_{sat}$, followed by combined horizontal-vertical (HV-type), while horizontally oriented roots (H-type) yield the lowest values, showing reductions of 11 %–32 % compared to bare soil.

(3) Plant roots influence soil $K_{sat}$ through two primary mechanisms. First, the development of channels at the root-soil interface creates preferential flow pathways, thereby increasing $K_{sat}$. Second, root expansion displaces and compresses surrounding soil particles, reducing porosity in the rhizosphere and consequently lowering $K_{sat}$. These dual effects interact to determine the overall hydraulic behaviour of vegetated soils.

(4) Novel void ratio functions have been developed to represent both the increased pore space due to root-soil interface channels and the reduced porosity caused by rhizosphere compaction. By integrating these functions with established models for predicting $K_{sat}$ of soil, a simplified yet effective predictive framework has been introduced. This model effectively captures the combined influence of root-induced soil compaction and preferential flow channel formation on the $K_{sat}$ of vegetated soils.

## CRediT authorship contribution statement

**Ling-Xin Cui:** Methodology, Validation, Writing – original draft, Investigation. **Qing Cheng:** Supervision, Conceptualization, Writing – review & editing, Funding acquisition, Methodology. **Chao-Sheng Tang:** Supervision, Writing – review & editing, Funding acquisition. **Pui San So:** Writing – review & editing. **Zhan-Ming Yang:** Investigation, Methodology. **Yang Lu:** Writing – review & editing. **Cong-Ying Li:** Investigation, Methodology. **Bin Shi:** Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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